

Digital Electronic Systems Design Project Report

Federico Ferro, Simone Guan, Xiao Chen Wang, Cheng Marco Zhang

Academic Year 2025/2026

Timing Analysis

From our implementation, the timing analysis yields compliant results for both setup and hold constraints at the target operating frequency of 150 MHz.

Regarding the setup time, the most critical path is located within the moving average block (Path 21). While the design requirement demands a 150 MHz clock, this implementation achieves a positive slack of 0.190 ns. This safety margin allows the circuit to potentially operate at a maximum frequency of approximately 154 MHz.

Concerning the hold time, the critical path is identified along the data route from the `diligent_jstk2` module to the `effect_selector_0` unit (Path 31). The required hold time constraint is 0 ns, and our implementation guarantees a safe positive hold slack of 0.026 ns, ensuring that no race conditions occur.

The detailed parameters resulting from the implementation are summarized in Table 1 for setup time and Table 2 for hold time below.

Tabel 1: Timing Summary: Setup Time (Path 21)

Parameter	Value	Unit
Slack	0.190	ns
Requirement	6.731	ns
Data Path Delay	6.519	ns
<i>Logic Delay</i>	2.916	ns
<i>Route Delay</i>	3.603	ns
Clock Path Skew	0.063	ns
Clock Uncertainty	0.116	ns

Tabel 2: Timing Summary: Hold Time (Path 31)

Parameter	Value	Unit
Slack	0.026	ns
Requirement	0.000	ns
Data Path Delay	0.360	ns
<i>Logic Delay</i>	0.141	ns
<i>Route Delay</i>	0.219	ns
Clock Path Skew	0.263	ns
Clock Uncertainty	0.116	ns

Power Analysis

At the operating frequency of 150 MHz, the following implementation's power consumption is as showed in the following tabels.

Tabel 3: Power Analysis Summary.

Parameter	Value	Unit
Total On-Chip Power	0.325	W
Design Power Budget	N/A	–
Power Budget Margin	N/A	–
Junction Temperature	26.6	°C
Thermal Margin	58.4	°C
Effective θ_{JA}	5.0	°C/W
Power to Off-Chip Devices	0	W
Confidence Level	Low	–

Tabel 4: On-Chip Power Breakdown.

Component	Power	Unit	Ratio (%)
Dynamic	0.252	W	77%
<i>MMCM</i>	0.125	W	50%
<i>BRAM</i>	0.077	W	30%
<i>Clocks</i>	0.017	W	7%
<i>I/O</i>	0.016	W	6%
<i>Signals</i>	0.010	W	4%
<i>Logic</i>	0.008	W	3%
<i>DSP</i>	0.000	W	0%
Device Static	0.074	W	23%